

João F. Melo

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PROFESSIONAL CAREER

Lecturer in Physics, Swansea University	Sep 2026 — Present
College Supervisor, University of Cambridge	Jan 2026 — Jun 2026
Postdoctoral Researcher, KU Leuven, Institute for Theoretical Physics	Jul 2023 — Jan 2025

HIGHER EDUCATION

PhD in Applied Mathematics and Theoretical Physics, DAMTP, University of Cambridge	Oct 2018 — May 2023
Thesis Title: The Many Scales of Quantum Fields and Gravity Supervisor: Jorge E. Santos	
Topics: Quantum Field Theory in Curved Spacetime, Renormalisation, Holography, Cosmology	
MAST in Applied Mathematics, DAMTP, University of Cambridge	Oct 2017 — Jun 2018
Result: Pass with Distinction, 90/100	
Exams Taken: Quantum Field Theory, Statistical Field Theory, String Theory, General Relativity, Black Holes	
Essay: T-Duality and Geometry of String Theory Advisor: Ron Reid-Edwards	
Graduate in Engineering Sciences – Engineering Physics, Instituto Superior Técnico, University of Lisbon	Sep 2014 — Jul 2017
Result: 18/20	
Core Topics: Theoretical Physics, Experimental Physics, Mathematics, Computing, Electrical Engineering	

PUBLICATIONS

1. Valeixo Bento, B. & Melo, J. F. EFT & Species Scale: Friends or foes? *JHEP* **25**, 212. arXiv: 2501.08230 [hep-th] (2025).
2. Melo, J. F. *The Many Scales of Quantum Fields and Gravity* PhD thesis (Cambridge U., May 2023).
3. Melo, J. F. The propagator matrix reloaded. *SciPost Phys. Core* **6**, 019. arXiv: 2112.09119 [hep-th] (2023).
4. Melo, J. F. & Santos, J. E. Stringy corrections to the entropy of electrically charged supersymmetric black holes with $\text{AdS}_5 \times S^5$ asymptotics. *Phys. Rev. D* **103**, 066008. arXiv: 2007.06582 [hep-th] (2021).
5. Melo, J. F. & Santos, J. E. Developing local RG: quantum RG and BFSS. *JHEP* **05**, 063. arXiv: 1910.09559 [hep-th] (2020).
6. Melo, J. F. Introduction to Renormalisation. arXiv: 1909.11099 [hep-th] (Sept. 2019).

TEACHING EXPERIENCE

College Supervisor, University of Cambridge	Jan 2026 — Present
• Courses: Applications of Quantum Mechanics, Electromagnetism, Statistical Physics	
• Instructor for small groups (1-3 undergraduate students) covering the problem sheets in the 2 nd and 3 rd year of the Mathematical Tripos	
Summer Camp Lecturer, AstroCamp, University of Porto	Aug 2025
• Course: Special Relativity: The Structure of Time and Space	
• Lecturer in an astrophysics summer camp (20 hours of lectures, 8 hours of practicals) for 10 school students from 15 to 17 years old	
• Wrote lecture notes, 2 problem sheets, and 3 exams (4 hours total)	
Master's Thesis Supervisor, KU Leuven	Sep 2024 — Dec 2024
• Thesis Title: Quantum Field Theory in Curved Spacetime	
• Co-supervisor for the early stages for a Master's Thesis in Physics	
Bachelor's Project Supervisor, KU Leuven	Sep 2024 — Dec 2024
• Project Title: Black Hole Thermodynamics	
• Sole supervisor of 2 students for their “capstone” project in the 3 rd year of the Bachelor of Physics	
• Their project included learning about General Relativity and performing symbolic calculations using Python	
Teaching Assistant, KU Leuven	Nov 2024
• Courses: Statistical Mechanics	
• Instructor for problem sessions for ~ 20 students in the 3 rd year of the Bachelor of Physics	
Lecturer, KU Leuven	Feb 2024 — Jun 2024
• Courses: Advanced Quantum Field Theory	
• Delivered one 2-hour lecture per week for 12 weeks for ~ 30 students in the Master of Physics	
• Wrote lecture notes (which can be found here) and 3 graded homework assignments	
• Conducted the final oral examination consisting of short presentations on a project written by each student chosen from a set list	
Examples Classes Instructor, University of Cambridge	Jan 2019 — Jan 2022
• Courses: General Relativity, String Theory	
• Sole instructor for ~ 20 master's students in the 4 th year of the Mathematical Tripos covering the problem sheets	

Part III Preparatory Workshops Instructor, University of Cambridge Oct 2019 — Oct 2021

- **Courses:** General Relativity
- Interactive workshop for ~ 50 master's students covering pre-requisites for the 4th year of the Mathematical Tripos
- Co-wrote notes with David Tong, these are still in use and can read [here](#).

PhD Drop-Ins, University of Cambridge Jan 2019 — Jun 2019

- **Courses:** General Relativity, String Theory, Statistical Field Theory
- Office hours for master's students in the 4th year of the Mathematical Tripos

ADMINISTRATIVE EXPERIENCE

Effective Field Theories Reading Group Oct 2020 — Dec 2020

Founder and Organiser

- Group for PhD students following the book “Effective Field Theories” by Andrew E. Blechman and Alexey A. Petrov

Racism Discussion Group Jun 2020 — Dec 2020

Co-Founder and Co-Organiser with Chandrima Ganguly, Hayley Macpherson, and Miren Radia

- Group for staff and students to discuss the issues of racial and ethnic discrimination in academia

Graduate Seminar Oct 2019 — Jun 2020

Co-organiser with Maeve Madigan

- Seminar for PhD students of the HEP and GR groups of DAMTP

Pre-Seminar Meetings Oct 2019 — Jun 2020

Founder and Co-Organiser with Filipe Miguel and Gonalo Regado

- Meetings between PhD students and the speaker of the Quantum Fields & Strings and General Relativity DAMTP seminars

Quantum Fields and Gravity Working Group Mar 2019 — Mar 2020

Founder and Organiser

- Group for PhD students at the intersection of the HEP and GR groups to share and discuss their research ideas

International Physics Olympiads Jun 2018

Liaison Officer

- Prepared and planned the daily activities for the students with the organising committee
- Coordinated a team of 10 guides each with their own delegation of 5 students
- With a team of 2 other liaison officers (each with their own 10 guide team) coordinated the full logistics of ~ 150 students. Including meals, transportation, electronics storage, and conflict resolution

OUTREACH

Demonstrating Experiments

Cambridge Hands-On Science Oct 2018 — Mar 2020

- **Target audience:** school students from 8 to 12 years old at schools and science festivals

NFIST - Physics Society of Instituto Superior Tcnico Sep 2014 — Aug 2017

- **Target audience:**
 - School students from 14 to 18 years old at schools, youth correctional centres, and science festivals
 - General audience at music festivals and on the street

Popular Science Writing

1. Clarke, P., Madigan M., **Melo, J. F.** Our Place in the Universe. *BlueSci*, 47 (2020)
2. **Melo, J. F.** Interview with David Tong (in Portuguese). *Pulsar*, 40 (2019)
3. **Melo, J. F.** Physics and Technology of the Telescope (in Portuguese). *Pulsar*, 38 (2017)
4. **Melo, J. F.** Physics and Technology of the Fridge (in Portuguese). *Pulsar*, 37 (2016)
5. **Melo, J. F.** Physics and Technology of the Television (in Portuguese). *Pulsar*, 36 (2016)
6. **Melo, J. F.** Physics and Technology of the Light Bulb (in Portuguese). *Pulsar*, 35 (2015)

SCHOLARSHIPS AND AWARDS

PhD Studentship, Fundao para a Cincia e Tecnologia Oct 2021 — Sep 2022

Smith-Knight and Rayleigh-Knight Prize, Faculty of Mathematics, University of Cambridge Jan 2020

Vice-Chancellor's Award, Cambridge Trust, University of Cambridge Oct 2018 — Sep 2021

Getrude Mather Jackson Prize, Girton College, University of Cambridge Jun 2018

M. T. Meyer Scholarship, Girton College, University of Cambridge Jun 2018

Merit Scholarship, Instituto Superior Tcnico, University of Lisbon Sep 2016 — Jul 2017

SHORT-TERM SCHOOLS

TASI, Black Holes, Quantum Information and Dualities Jun 2021

AQFTUK/LMS, An Introduction to Quantum Field Theory in Curved Spacetime Oct 2020

CERN, Supergravity, Strings and Gauge Theory Feb 2019

LIP/CFTP, Particle and Astroparticle Physics Feb 2017

University of Coimbra, Projecto Quark! – Preparation for the International Physics Olympiads Jan 2014 — Jun 2014